

Lecture 10: Exchange-Rate Policy and Central Bank

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Overview

This lecture studies the link between a country's exchange rate policy and its domestic monetary policy.

Key insight:

- Central banks must choose between a fixed exchange rate and an independent monetary policy.
- Trilemma of international finance: A country cannot simultaneously [i.] maintain a fixed exchange rate; [ii.] conduct independent monetary policy; [iii.] allow free capital flows

The Road Ahead

1. Exchange-Rate Policy and Domestic Monetary Policy
2. Mechanics of Exchange-Rate Management

Long-Run: Inflation and PPP

The theory of purchasing power parity (PPP) extends the law of one price to a basket of goods and services. Taking percentage changes of the PPP condition:

$$\% \text{ change in } E \approx \underbrace{\% \text{ change in } P^*}_{\text{U.K. inflation}} - \underbrace{\% \text{ change in } P}_{\text{U.S. inflation}}$$

Implication: Pound depreciates (appreciates) when U.K. inflation exceeds (falls below) U.S. inflation.

Long-run constraint: Bank of England must choose between

- Fixing the exchange rate and matching U.S. inflation
- Setting independent monetary policy and allowing the exchange rate to float

Short-Run: Interest Rates and Capital Arbitrage

Arbitrage in capital markets ensures equal expected returns on U.S. and U.K. bonds:

$$1 + i_t = (1 + i_t^*) \left(\frac{E_t}{E_{t+1}^e} \right)$$

Under a credible fixed exchange rate ($E_t = E_{t+1}^e$), this implies $i_t = i_t^*$.

Short-run constraint: Bank of England must choose between

- Fixing the exchange rate and matching U.S. interest rates
- Setting independent monetary policy and allowing the exchange rate to float

Impossible Trinity

If a country forgoes participating in international capital markets, it can impose **capital controls** and potentially achieve both goals. Policymakers must choose two of the three:

1. Be open to international capital flows
2. Control domestic interest rate
3. Fix exchange rate

Policy choices in practice:

- Large economies (U.S., Eurozone) prioritize free capital flows and monetary independence, letting exchange rates float.
- Small open economies may prioritize exchange rate stability given dramatic impacts of exchange rate changes.

Mechanics of Exchange-Rate Management

How do central banks actually manage exchange rates?

Mechanism: Central banks can fix exchange rates by offering to buy and sell their country's currency at the fixed rate.

Implication: Controlling the exchange rate means giving up control of the size of reserves, so the market determines the interest rate.

Foreign Exchange Intervention

Suppose the U.S. Treasury instructs the Fed to purchase \$1 billion worth of euros.

Mechanism:

- Fed buys German government bonds (denominated in euros)
- Pays foreign exchange departments of commercial banks with dollars

Impact on Fed's balance sheet:

- Assets increase by \$1 billion (foreign exchange reserves)
- Liabilities increase by \$1 billion (bank reserves)
- Monetary base increases by \$1 billion

Figure 17.3

Balance Sheet Changes after the Federal Reserve Purchases a German Government Bond

A. Federal Reserve's Balance Sheet		B. Banking System's Balance Sheet	
Assets	Liabilities	Assets	Liabilities
Foreign exchange reserves +\$1 billion (German government bonds in euros)	Reserves +\$1 billion	Reserves +\$1 billion	
		Securities -\$1 billion (German government bonds)	

Effects on Money Supply and Interest Rates

Through money multiplier:

- Monetary base increases
- Quantity of money in economy increases
- Supply of reserves to banking system increases

Effect on interest rates:

- Increased reserve supply puts downward pressure on interest rates
- Buying euros (selling dollars) lowers U.S. interest rates

This is an **unsterilized** foreign exchange intervention (changes the monetary base).

Impact on Exchange Rate

Figure 19.2

Effect of a Decrease in U.S. Interest Rates Relative to Interest Rates in the Euro Area

